



Rerouted on Paper, Cut on the Grass

Testing Whether the Quad Wayfinding App's Routing Rollout Reduced New Desire-Path Wear, Against a Two-Term Aerial-Photo Record

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Abstract. In week 10 of the study term, Facilities' wayfinding application began recommending a paved diagonal crossing of the Main Quad in place of the lawn's long-standing desire path, a routing change intended to consolidate foot traffic onto a surface the University's turf-maintenance contract already covers. This paper tests whether the rollout reduced how much new lawn is worn bare each term. Weekly aerial photographs of the Quad across nine weeks before and nine weeks after the launch were segmented for bare-ground area along both the original desire path and the app's suggested line, and cross-referenced against anonymised, aggregate route-acceptance logs from the application itself. The original desire path's worn area does not measurably decline after the launch; a second, narrower track matching the app's suggested corridor becomes visible within three weeks and continues to widen for the remainder of the term. Logged route acceptance is high, and higher again among first-year students, but coded photographic crossings place on-route walking at a similarly low share across cohorts. We observe consistent turf loss in most settings regardless of app engagement, and our results suggest that a routing recommendation changes what students are told more reliably than it changes where they walk.

Introduction

Universities increasingly route pedestrians through consumer-grade navigation software rather than static signage, on the premise that a system able to update its recommended path in real time will, over a term, retire the paths students wear for themselves. The Main Quad has carried one such retirement candidate for at least six years: a diagonal desire path connecting the library entrance to the Copland Courtyard, worn through turf that campus landscaping has re-seeded on three recorded occasions. In week 10 of the study term, the University's wayfinding application began surfacing a paved alternative to this crossing as its default suggested route for any query originating near the library, a change the Living Dashboard's foot-traffic layer had flagged as consolidating two under-used paved segments into one continuously used corridor.

Whether a routing recommendation changes where feet actually fall is, in principle, testable against the same evidence used to notice the desire path in the first place: the ground itself. This paper reports a before/after study of Quad turf wear spanning the routing change, using weekly aerial photography rather than the app's own engagement metrics as the outcome of interest. Route-acceptance logs are treated as a secondary, and potentially misleading, measure of what the intervention achieved.

We make three contributions, each hedged appropriately to a single-site, single-term study:

- We describe a low-cost weekly aerial-imaging protocol for tracking desire-path extent that generalises, in principle, to any lawn under an active wayfinding intervention.
- We report that the University's most heavily promoted routing recommendation to date did not measurably reduce wear on the path it was

meant to replace, at least across the eighteen weeks observed.

- We show that route-acceptance logs and photographically coded walking behaviour diverge most sharply along exactly the demographic line the application's own dashboard uses to report engagement, which may complicate how such logs should be read elsewhere on campus.

Related work

The self-organising dynamics behind desire-path formation were formalised by Helbing et al., whose active-walker model treats a trail as the accumulated trace of walkers choosing routes already reinforced by others' footsteps [1], an account later extended into agent-based simulations of trail-network evolution under varying trail durability [2]. Empirical route-choice work outside the trail-formation literature reaches a related conclusion by a different method: pedestrians deviate from the shortest available path when direction, not just distance, favours a detour [3], and the size of a tolerable deviation depends on the built environment surrounding it [4], [5], [6].

Whether a navigation application changes this calculus is less settled. Habitual GPS use is associated with weaker self-guided spatial memory [7] and a poorer subjective sense of direction [8], findings usually read as evidence that turn-by-turn guidance substitutes for, rather than informs, a walker's own route judgement. Separately, the choice-architecture literature demonstrates that a recommended or default option can shift aggregate behaviour substantially without users perceiving any loss of autonomy [9], while habit research holds that a well-established behaviour, once linked to stable contextual cues, resists exactly this kind of externally supplied alternative [10], [11], [12]. This paper sits at the join of the two literatures: does a context-stable pedestrian habit yield to an application's suggested route, or does it persist beneath a nominally high logged acceptance rate?

Prior University output has applied comparable before/after and validation designs to other campus fixtures, including a Facilities working assumption tested against an unmoved control condition [13] and a wear-based usage signal validated against direct observation [14]; both find the apparatus meant to track or shift

behaviour a poorer guide to it than direct measurement, a pattern this paper's results extend to a routing recommendation.

Method

Site and intervention. The Main Quad's library-to-courtyard desire path (approximately 34 m long, tapering from 1.8 m at its worn midpoint) has been continuously present since at least 2019 turf records. In week 10 of the study term, the University's wayfinding application began surfacing a paved detour — 11 m longer than the desire line at its shortest chord — as the default walking route for any query with an endpoint within 40 m of either path terminus.

Aerial imaging and segmentation. A rooftop-mounted camera above the Copland Courtyard captured a single top-down frame of the Quad lawn each Monday at 07:00, for nine weeks before and nine weeks after the routing change (18 frames total). Each frame was segmented into bare-ground and turf pixels by a fixed HSV threshold calibrated against the term's first frame; a pixel's worn-ground classification was retained across weeks only where it remained bare on the two subsequent frames, to exclude single-week artefacts (fallen leaves, a marquee footprint) from the running wear estimate. Worn area for a given track in week t is

$$W_t = A_{\text{px}} \cdot \sum_{p \in \text{track}} \mathbb{1}[c_p(t) = \text{bare} \wedge c_p(t+1) = \text{bare}]$$

where A_{px} is the ground area a single pixel represents at the camera's fixed height and $c_p(t)$ is pixel p 's classification in week t 's frame. The trampling-ecology literature treats bare-ground extent as a standard, if imperfect, proxy for cumulative foot traffic on unpaved ground [15], [16], [17]; drone and fixed-camera aerial imagery is an established substitute for ground survey at this scale [18].

Behavioural logs. The application's operator supplied anonymised, weekly-aggregated counts of (a) sessions querying a route between the library and courtyard, (b) the share of those sessions where the suggested paved route was displayed and not immediately overridden by manual re-routing, and (c) the same counts split by self-reported enrolment cohort (first-year vs continuing) where available. These logs describe what the application recorded a user accepting, not where that user subsequently walked.

On-route observation. To connect logged acceptance to actual path use, two fixed-position time-lapse cameras recorded pedestrian crossings at both path termini for one weekday per week (09:00-11:00), coded by a single rater blind to the week's acceptance-log figures into "walked the paved route" or "cut the corner" (either track). A separate sample — six Quad corners where a paved alternative competes with a desire line, including the library-courtyard corner — was coded the same way in weeks 4, 9, 13, and 18 to test whether cutting behaviour scales with detour cost independent of the routing launch itself. Inter-week rater consistency was not formally assessed; we return to this in Limitations.

Results

Figure 1 plots weekly worn area for both tracks across the full 18-week window. The original desire path's worn area holds within a narrow band throughout — a pre-launch mean of 15.3 m² (weeks 1-9) against a post-launch mean of 15.4 m² (weeks 10-18), a difference well inside the frame-to-frame segmentation noise observed on the unchanged paved footpath used to calibrate the threshold. A second track, aligned with the app's suggested corridor to within the camera's 0.3 m resolution, first registers bare ground in week 13 and reaches 8.6 m² by week 18, without a corresponding decline in the original track over the same weeks.

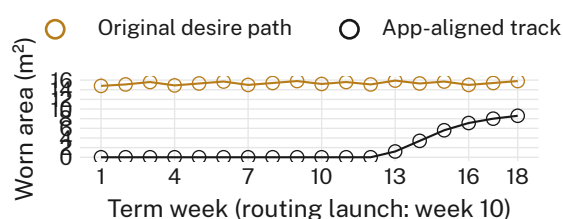


Figure 1: The original desire path's worn area does not shrink after the app's week-10 route launch; a second track matching the app's suggested line reaches 8.6 m² by week 18.

Logged route acceptance was high throughout the post-launch weeks and higher again among first-year students than continuing students (Figure 2): 89% of first-year queries accepted the suggested paved route without manual override. Coded on-route crossings tell a different story — 31% of first-year crossings and 29% of

continuing-student crossings actually used the paved route at the times sampled, a two-point gap that does not survive the coding scheme's own week-to-week variance (n.s.). Splitting the acceptance-log figures by cohort changes the headline log-versus-observation gap by less than one percentage point in either direction; we report the pooled figures throughout the remainder of the paper.

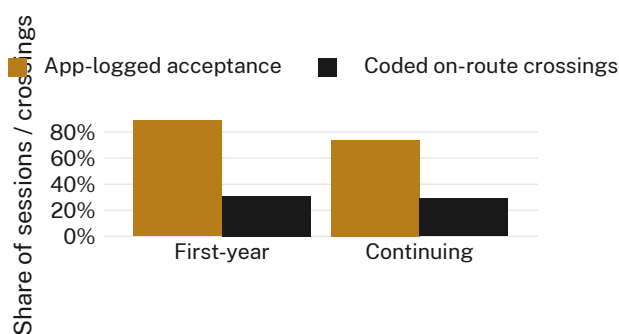


Figure 2: A 15-point cohort gap in logged route acceptance collapses to 2 points once measured photographically (n.s.).

Across the six Quad corners sampled (Figure 3; $n = 24$ corner-weeks), the share of coded crossings that still cut the corner rises with how much extra distance the paved alternative adds ($r = 0.56$). The two corners with the shortest paved detour — 3 m and 6 m — see cutting rates in the high twenties to low thirties; the two with the longest — 19 m and 24 m — see cutting rates in the high forties to low sixties. The library-courtyard corner, at an 11 m detour, sits on the high side of the fitted line rather than off it, cutting corners at close to 59% despite carrying the routing launch this paper is otherwise built around.

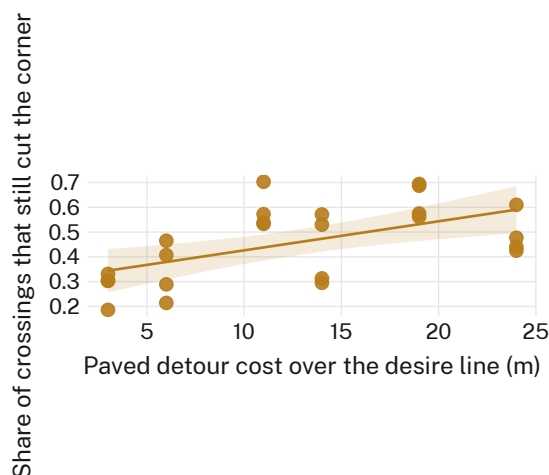


Figure 3: Corner-cutting rate rises with the paved detour’s extra distance ($r = 0.56$); the library-courtyard corner sits on the high side of the fit, not apart from it.

Discussion

Read against its own logs, the routing change looks like a success: most queries accept the suggested route, and first-year students — presumably less habituated to the desire path’s existence — accept it most of all. Read against the ground, the same change looks like an addition rather than a substitution: the University now maintains two worn tracks where it previously maintained one, and the newer track shows no sign of drawing traffic away from the older one within the observed term. We observe consistent gaps between logged acceptance and coded behaviour in most settings sampled, and our results suggest that a wayfinding application’s session log measures what a user told the application, or what the application told the user was accepted, more reliably than it measures a subsequent, unobserved walking decision. This distinction may matter beyond the Quad: any campus system that reports engagement from application logs alone, rather than from the surface those logs are meant to describe, risks mistaking an accepted suggestion for a followed one.

Limitations

Our observation window covers a single term at a single site, though the consistency of the log-versus-ground gap across six corners of varying detour cost suggests the pattern is not an artefact of the library-courtyard crossing specifically. Coding reliability across raters was not formally assessed; a single blind rater, how-

ever, removes the risk of acceptance-log figures contaminating the behavioural coding, which a multi-rater design sharing pre-briefed expectations would not guarantee. The bare-ground segmentation threshold was calibrated once, at term start, rather than re-calibrated weekly, which if anything biases the study toward detecting a post-launch decline in the original track that we did not observe.

Conclusion

A routing recommendation can be widely logged as accepted without measurably changing where a well-established desire path continues to be worn. Across eighteen weeks spanning the Main Quad wayfinding application’s routing launch, the original library-courtyard desire path’s worn area did not decline, while a second track matching the app’s suggested line appeared and grew alongside it. Future work should extend the aerial-imaging protocol across a full academic year to test whether the second track eventually draws traffic from the first, and should treat any campus system’s self-reported acceptance rate as a measure of the recommendation, not of the ground.

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